

Spring-Loaded Piezoelectric Generating Striker Device: Design, Methodology, and Applications

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ABSTRACT

This paper presents the design, working mechanism, and potential applications of a novel Spring-Loaded Piezoelectric Generating Striker Device. The device integrates a piezoelectric transducer, a spring-driven hammer mechanism, and an energy storage module to convert ambient mechanical vibrations and footstep impacts into usable electrical energy. When an external force is applied, the spring compresses and releases, propelling a hammer to repeatedly strike the piezoelectric surface and generate an electric charge. The compact, durable design makes it highly suitable for deployment in high-footfall environments such as pedestrian walkways, industrial floors, and smart infrastructure nodes. The novelty of the device lies in its distinctive shape and configuration, for which a Design Registration application (Form-1) has been submitted to the Controller of Designs, The Patent Office, Kolkata, India. This paper explores the theoretical underpinnings of piezoelectric energy harvesting, the mechanical design parameters, and the practical pathways to large-scale renewable energy generation.

Keywords: Piezoelectric Energy Harvesting, Spring Mechanism, Kinetic-to-Electrical Conversion, Footstep Energy, Sustainable Energy, Smart Infrastructure, Renewable Power

1. Introduction and Background

The global demand for clean, distributed, and renewable energy sources has intensified research into ambient energy harvesting — the process of capturing small amounts of energy that exist in the environment and converting them into usable electricity. Among the most promising approaches is piezoelectric energy harvesting, which exploits the piezoelectric effect: the ability of certain crystalline and ceramic materials to generate an electrical potential in response to applied mechanical stress.

Conventional large-scale renewable sources such as solar panels and wind turbines require significant infrastructure investment and are subject to environmental intermittency. By contrast, piezoelectric harvesters are compact, passive, and capable of generating continuous power from ubiquitous mechanical inputs — vibrations from machinery, vehicle traffic, ocean waves, and, most relevantly,

human foot traffic. High-density pedestrian areas such as train stations, shopping malls, airports, and school corridors represent an enormous untapped reservoir of mechanical energy.

The Spring-Loaded Piezoelectric Generating Striker Device was conceived at Charotar English Medium School, Anand, Gujarat, by a team of student inventors under faculty mentorship. The device addresses a fundamental challenge in piezoelectric harvesting: maximising the impact force delivered to the piezoelectric element. By incorporating a spring-loaded hammer, the device amplifies the mechanical impulse transmitted to the piezoelectric surface, significantly increasing the electrical output per activation cycle compared to direct-pressure designs.

A Design Registration application under Sections 5 and 44 of the Designs Act (Form-1) was submitted to the Patent Office, Kolkata on 27 February 2025, claiming novelty in the shape and configuration of the device. This paper elaborates on the technical design, working principles, and the broad spectrum of applications enabled by this invention.

2. Design and Methodology

2.1 Core Components

The device comprises three principal sub-systems, each optimised for its functional role within the energy conversion chain:

Sub-System	Component	Function
Mechanical Input	Pressure Platform	Receives foot/mechanical load; transmits force to the spring.
Energy Amplifier	Spring-Loaded Hammer	Stores elastic potential energy; releases it as a rapid kinetic impulse on the piezoelectric surface.
Transducer	Piezoelectric Crystal / PZT Disc	Converts mechanical stress into an alternating electric charge via the direct piezoelectric effect.
Energy Management	Rectifier + Storage Module	Converts AC output to DC; stores energy in a capacitor or rechargeable battery for end-use.

Table 1: Principal sub-systems and their functions.

2.2 Working Principle

The operational cycle of the device proceeds through four distinct phases:

Step 1 – Loading Phase: An external force — typically a footstep — depresses the pressure platform. This compresses the internal spring, storing mechanical (elastic potential) energy proportional to the spring constant k and displacement x : $E = \frac{1}{2}kx^2$.

Step 2 – Release Phase: Upon reaching the compression threshold or when the load is removed, the spring releases stored energy and propels the hammer head at high velocity toward the piezoelectric disc. The rapid, controlled impact maximises the stress applied to the piezoelectric material.

Step 3 – Transduction Phase: The hammer strikes the piezoelectric element, inducing mechanical deformation. The piezoelectric effect (direct effect, described by $d = P/T$ where P is polarisation and T is mechanical stress) causes charge separation within the crystal lattice, generating an alternating voltage across the electrode faces of the disc.

Step 4 – Energy Storage Phase: The generated AC voltage is passed through a bridge rectifier to produce DC, which is smoothed and stored in a capacitor bank or lithium-ion battery. The stored energy is then available to power low-energy electronic devices or to be fed into a micro-grid.

2.3 Design Configuration and Views

The device exhibits a distinctive shape that is central to its registered design claim. Seven engineering views were prepared and submitted as part of the design registration:

Figure	View	Key Observable Feature
1	Front View	Angled support struts, central spring-hammer column, flat pressure platform, and base chassis with cyan accent strip.
2	Rear View	Dual output ports (circular connectors) for electrical lead-out; symmetrical rear profile.
3	Top View	Circular platform with concentric gold-coloured piezoelectric ring and dark central impact zone.
4	Bottom View	Octagonal base plate with four corner mounting points for surface installation.
5	Left Side View	Profile of strut angle, vertical spring column height, and platform overhang.
6	Right Side View	Mirror profile confirming bilateral symmetry of the structural frame.
7	Perspective View	Three-dimensional isometric rendering showing the overall form factor and assembly integration.

Table 2: Engineering views submitted with the design registration (Sheets 1–7 of 7).

2.4 Optimisation Parameters

The electrical output of the device is governed by several interacting mechanical and material parameters. Key variables available for engineering optimisation include:

- **Spring Stiffness (k):** A stiffer spring stores more energy per unit displacement but requires greater input force. Optimum k must balance energy storage with the typical force profile of the intended application.
- **Hammer Mass and Geometry:** A heavier hammer delivers greater impact impulse; the contact geometry (flat vs. convex) affects the stress distribution across the piezoelectric disc and, consequently, the charge generated.

- **Piezoelectric Material Selection:** Lead Zirconate Titanate (PZT-5A, PZT-5H) offers high piezoelectric coefficients ($d_{33} \sim 400\text{--}600 \text{ pC/N}$), making it the preferred material for high-output applications. PVDF films are a flexible, lightweight alternative for integration into textiles or curved surfaces.
- **Electrode Coverage and Wiring:** Maximising the effective electrode area and using series/parallel stacking of multiple piezoelectric discs can multiply voltage or current output respectively, depending on the load requirement.
- **Striker Frequency:** Higher activation frequency (more footsteps per unit time) increases average power output. Device placement in high-footfall corridors directly influences this parameter.

3. Applications and Use Cases

The versatility of the spring-loaded piezoelectric striker design enables deployment across a wide range of sectors. The following sub-sections detail the primary application domains.

3.1 Footstep Energy Harvesting in Public Infrastructure

This is the most direct and immediate application. Arrays of the device embedded beneath floor tiles or pavement slabs in high-footfall locations — railway stations, airport terminals, shopping malls, university campuses, and public parks — can generate sufficient power to operate LED lighting, digital signage, Wi-Fi access points, and environmental sensors. A single pedestrian generates approximately 1–5 W of power per footstep with a well-designed harvester; a corridor with 10,000 daily crossings could sustain continuous low-power operations without grid dependency.

3.2 Industrial Machinery and Structural Health Monitoring

Industrial floors, bridge decks, and machinery housings are subject to continuous vibration. Mounting the striker device at vibration hotspots allows it to self-power wireless sensor nodes that monitor structural integrity, equipment health, and environmental parameters. This eliminates the need for periodic battery replacement in remote or hazardous locations — a critical advantage for predictive maintenance systems in the manufacturing and civil engineering sectors.

3.3 Smart Infrastructure and IoT Power Supply

The Internet of Things (IoT) ecosystem requires billions of low-power nodes — smart meters, environmental monitors, traffic counters, and security devices — that are costly to maintain with conventional batteries. The spring-loaded piezoelectric striker device provides a self-sustaining power source for such nodes when positioned at points of regular mechanical activity. Integration with energy-aware microcontrollers (e.g., ultra-low-power ARM Cortex-M series) allows harvested energy to be duty-cycled efficiently, supporting continuous operation from intermittent mechanical inputs.

3.4 Military and Defence Applications

In field deployments where grid power and battery logistics are constrained, piezoelectric harvesters embedded in soldier boots or vehicle floor panels can power personal area network devices, GPS modules, and physiological monitoring sensors. The device's robustness — stemming from its

all-mechanical energy storage (spring) and solid-state transduction (piezoelectric disc) — makes it well-suited for harsh operational environments.

3.5 Wearable and Biomedical Devices

Miniaturised variants of the spring-hammer principle can be integrated into orthopaedic insoles, prosthetic limbs, or implantable biomedical devices. Harvesting energy from joint movement or cardiac contractile force provides a promising pathway toward self-powered pacemakers, neural stimulators, and continuous glucose monitoring systems — reducing patient risk from battery-replacement surgeries.

3.6 Educational and Research Demonstrations

As a school-originated innovation, the device naturally lends itself to STEM education. Its transparent operating principle — compress a spring, strike a crystal, light an LED — makes abstract energy conversion concepts concrete and engaging for students at all levels. Schools and science centres can deploy units as interactive exhibits, simultaneously generating awareness of renewable energy and inspiring the next generation of engineers.

Application Domain	Power Scale	Primary Benefit
Public Footpaths / Floors	mW – W	Grid-free lighting and signage
Industrial Vibration Harvesting	mW – W	Self-powered structural sensors
IoT Node Power Supply	μ W – mW	Battery-free smart devices
Military / Field Operations	mW	Logistics-independent power
Wearable / Biomedical	μ W – mW	Implant / wearable self-powering
STEM Education	μ W – mW	Concept demonstration and outreach

Table 3: Summary of application domains, indicative power scales, and primary benefits.

4. Conclusion and Future Scope

4.1 Conclusion

The Spring-Loaded Piezoelectric Generating Striker Device represents a significant conceptual advance in ambient mechanical energy harvesting. By decoupling the energy input event (footstep, vibration) from the striker impact event through a spring intermediary, the device overcomes a key limitation of direct-pressure piezoelectric harvesters — the dependence of output voltage on the speed of the applied force. The spring mechanism stores energy regardless of input speed and releases it at a consistently high rate, ensuring a reliable and repeatable electrical output.

The design's novelty — recognised through its formal Design Registration filing in India — lies in its unique shape and structural configuration: the circular piezoelectric platform supported by an angled strut frame, the central spring-hammer column, and the octagonal mounting base. This configuration optimises mechanical force transmission while maintaining a compact form factor suitable for seamless

integration into standard floor tile formats.

The multi-disciplinary team of student inventors, guided by experienced faculty mentors at Charotar English Medium School, has demonstrated that impactful innovation in sustainable energy is not confined to well-funded research laboratories. This project stands as proof of the transformative potential of school-level STEM programmes.

4.2 Future Scope

- **Prototype Fabrication and Testing:** Constructing physical prototypes to measure open-circuit voltage, short-circuit current, power output, and efficiency under controlled and real-world footstep conditions. Comparison with simulation models to validate design parameters.
- **Material Optimisation:** Exploring advanced piezoelectric composites such as PMN-PT single crystals ($d_{33} > 2,000$ pC/N), flexible PVDF-TrFE films, and bio-inspired piezoelectric materials to improve transduction efficiency and reduce environmental impact.
- **Multi-Unit Array Deployment:** Designing tile-format arrays of multiple striker units that can be modularly connected in series (higher voltage) or parallel (higher current) configurations. Field testing in school corridors and public spaces to measure cumulative energy generation.
- **Integrated Energy Management System:** Developing dedicated power conditioning circuits with maximum power point tracking (MPPT), ultra-low-quiescent-current rectifiers, and energy-aware microcontrollers to maximise usable output from harvested energy.
- **Hybrid Harvesting:** Combining piezoelectric elements with triboelectric nanogenerators (TENG) or electromagnetic induction coils within the same chassis to harvest complementary frequency ranges of mechanical input, increasing overall power density.
- **IP Development and Commercialisation:** Progressing from Design Registration to full Patent protection under the Indian Patents Act; exploring licensing partnerships with infrastructure developers, smart city projects, and IoT hardware manufacturers.
- **Life-Cycle Assessment:** Conducting environmental impact analyses to quantify the net carbon benefit of the device over its operational lifetime, accounting for manufacturing energy, material sourcing, and end-of-life recyclability.

In summary, the Spring-Loaded Piezoelectric Generating Striker Device has the potential to contribute meaningfully to India's renewable energy targets and global sustainability goals. With appropriate investment in prototyping, materials research, and system integration, this student-conceived invention could transition from a registered design to a commercially deployed product within the next few years.

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References

- [1] Anton, S. R., & Sodano, H. A. (2007). A review of power harvesting using piezoelectric materials (2003–2006). *Smart Materials and Structures*, 16(3), R1–R21.
- [2] Beeby, S. P., Tudor, M. J., & White, N. M. (2006). Energy harvesting vibration sources for microsystems applications. *Measurement Science and Technology*, 17(12), R175–R195.
- [3] Erturk, A., & Inman, D. J. (2011). *Piezoelectric Energy Harvesting*. John Wiley & Sons.
- [4] Priya, S., & Inman, D. J. (Eds.). (2009). *Energy Harvesting Technologies*. Springer.
- [5] Roundy, S., Wright, P. K., & Rabaey, J. (2003). A study of low-level vibrations as a power source for wireless sensor nodes. *Computer Communications*, 26(11), 1131–1144.
- [6] Wang, Z. L., & Song, J. (2006). Piezoelectric nanogenerators based on zinc oxide nanowire arrays. *Science*, 312(5771), 242–246.
- [7] Designs Act, 2000 (India). Design Registration Form-1 (Application for Registration of Designs) under Sections 5 and 44. The Patent Office, Kolkata.
- [8] Charotar English Medium School Student Inventors. (2025). *Design Registration Application — Spring-Loaded Piezoelectric Generating Striker Device*. Filed 27 February 2025, The Patent Office, Kolkata, India.